

Thermometric scales

Absolute zero, and what a thermometer really does

Physics · 3A · 11 September 2026

Where we left off

(slide 2–3)

Temperature is not a sensation. It is a label. We do not restart from zero on equilibrium: we built it on 8 September and today we use it. Three things from there.

- Your hand does not measure temperature: it measures how fast energy leaves your finger.
- “Being in thermal equilibrium with” is an equivalence relation: reflexive, symmetric and transitive. The transitive part is the Zeroth Law.
- Temperature is the name of an equivalence class: two systems have the same temperature exactly when they are in the same class, that is, in thermal equilibrium with each other.

Today we ask what number to attach to a class, and how to build the instrument that reads it.

One sentence for the whole term

(slide 4)

“This body has a lot of heat” is exactly as wrong as “this body has a lot of work”.

- Temperature is a **state** quantity: a property a system has right now, like mass or volume.
- Heat is a **process** quantity: it only exists while energy is being transferred because of a temperature difference. Once the transfer stops, there is no more “heat” to speak of.
- What the body actually has, at any instant, is **internal energy**: the total energy of all its particles.

The Celsius scale: two fixed points

(slide 5–7)

0 °C: the melting point of pure ice. 100 °C: the boiling point of pure water, at standard atmospheric pressure.

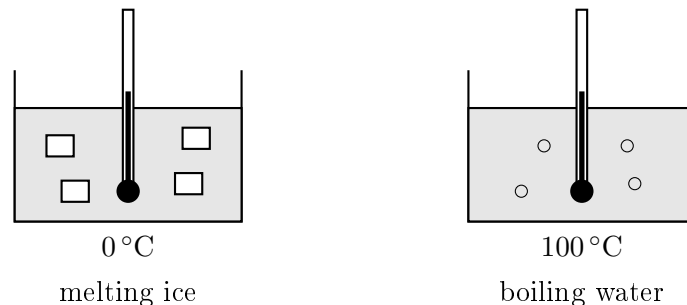


Figure 1: The two fixed points of the Celsius scale: melting ice (0 °C) and boiling water (100 °C), at standard atmospheric pressure (schematic).

The phrase “at standard atmospheric pressure” matters: at different altitudes water boils at different temperatures.

How cold can something actually get?

(slide 8)

Do not take my word for it: let us find out. A fixed amount of gas is kept at constant pressure. Its volume is measured at four different temperatures.

Temperature / °C	0	50	100	150
Volume / cm ³	100	118	137	155

The steps between temperatures are equal (50 °C each time). Look at how the volume changes at each step.

Before moving on. If this pattern kept going in a straight line, with no phase changes and nothing weird, at what temperature would the volume reach zero? For a first estimate you do not need graph paper.

The reveal

(slide 9)

Every 50°C the volume grows by about 18 cm^3 (18, 19, 18). On average $(155-100)/3 \approx 18.3\text{ cm}^3$ per 50°C , so to bring the volume from 100 down to 0 takes $100/18.3 \approx 5.5$ steps of 50°C : about 273°C *below* zero (figure 2; the line of best fit through the four points gives about -271°C , in agreement within the scatter).

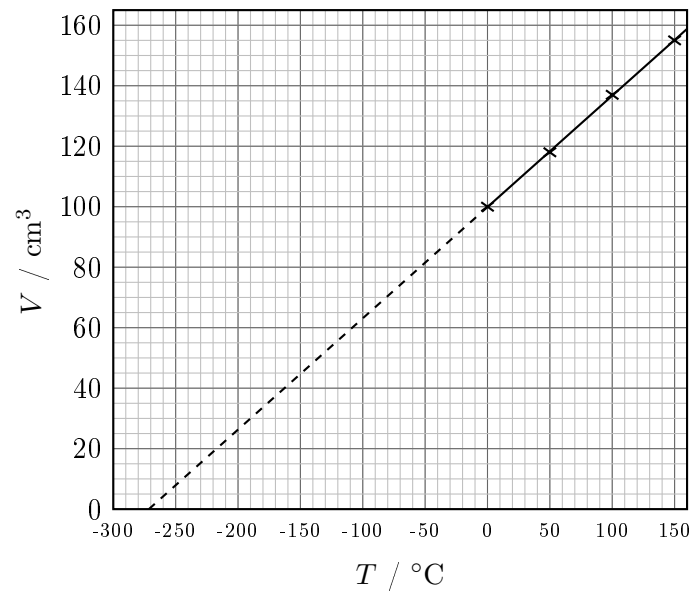


Figure 2: Volume of a gas at constant pressure against temperature: the four measured points (crosses), the line of best fit and its extension (dashed) down to $V = 0$.

Whatever gas you use, the line crosses zero at the same point. This is not a rule someone invented: it is a temperature that experiments kept pointing to, long before anyone could reach it. Historically this is how absolute zero was discovered, by extrapolating the pressure or volume of a gas towards zero (Gay-Lussac, then Kelvin).

$$T/\text{K} = T/^\circ\text{C} + 273$$

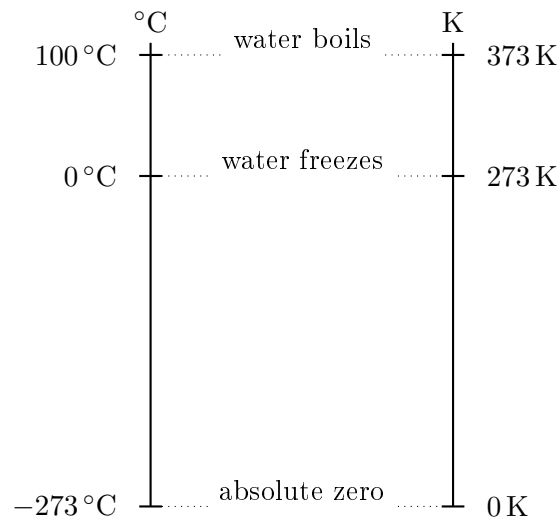


Figure 3: The Celsius and Kelvin scales side by side: $T/\text{K} = T/^{\circ}\text{C} + 273$. Absolute zero is -273°C , that is 0 K.

The check that matters

(slide 10)

Before moving on. Why is there no such thing as -50 K ?

If your answer is “because that is the rule”, the concept has not landed yet. The physical answer: the kelvin is built *on top of* the physical fact that there is a coldest possible state. It is not the scale that forbids negative numbers: it is reality.

What actually happens near absolute zero?

(slide 11–12)

Matter starts breaking its own rules. Three real, measured phenomena.

- **Superconductivity:** some materials lose *all* electrical resistance. A current started in a loop of superconducting wire can keep flowing, essentially forever, with nothing lost as heat.
- **Superfluidity:** liquid helium flows with zero viscosity. It can climb up and over the wall of its own container, against gravity, just to reach a lower level outside.
- **Bose–Einstein condensate:** cool a cloud of atoms enough and they lose their individual identity, merging into one single giant quantum wave, as if they were one atom.

All three have been produced and measured in real laboratories, including at CERN.

Where is the coldest place in the known universe?

(slide 14–17)

Before moving on. Where is the coldest place in the known universe? Guess before you read on.

It depends what you mean by “place”.

- The coldest large, human-built region we know of: CERN’s Large Hadron Collider, 1.9 K (-271.3°C), colder than the vacuum of outer space itself (2.7 K, the leftover glow of the Big Bang).
- Colder still, but natural and 5000 light-years away: the Boomerang Nebula, about 1 K, gas expanding so fast it cools itself down.
- Colder than either, but only in a speck a few atoms wide: laboratory experiments have reached temperatures thousands of times closer to zero than deep space.

Biggest engineered cold region: CERN. Coldest natural spot: a nebula. Coldest ever, at any size: a physics lab.

-271.3°C corresponds to 1.9 K with the exact conversion 273.15; with 273 one gets -271.1°C .

Does time stop at absolute zero?

(slide 18–19)

A popular myth: let us check it.

Before moving on. Does time stop at absolute zero?

No, and the reason is more interesting than the myth. The myth pictures 0 K as “all motion completely stops”: that part is already false. Quantum mechanics (Heisenberg’s uncertainty principle) guarantees a tiny irreducible motion even at 0 K, called *zero-point energy*: a particle can never be perfectly still *and* perfectly located at the same time. Time itself is not a thermal quantity: it does not slow down because something got cold. What grinds to a halt near 0 K is chemistry: reactions become so slow they might as well not happen.

The real fact, better than the myth: nothing, anywhere, is ever perfectly motionless.

Two conversions, worked together

(slide 20)

Room temperature, about 20°C : $20 + 273 = 293\text{ K}$.

Liquid nitrogen boils at 77 K: $77 - 273 = -196^{\circ}\text{C}$.

The five conversions in your homework use exactly this: no new rule, just practice.

The Fahrenheit scale: which two points, exactly?

(slide 21)

- 0 °F: the coldest temperature Fahrenheit could reach with a mixture of ice, water and salt.
- 96 °F: human body temperature, as he measured it. Water freezing and boiling were not the plan: they just came out at 32 °F and 212 °F.
- Later scientists fixed 32 °F and 212 °F as the two official reference points, exactly 180 degrees apart: body temperature then had to be recalculated to 98.6 °F.
- It is still the everyday scale in the United States and a handful of other places.

$$T/^{\circ}\text{F} = T/^{\circ}\text{C} \times \frac{9}{5} + 32$$

−40 °F and −40 °C are the same temperature: the one point where the two scales agree.

Celsius and Fahrenheit were never the only two

(slide 22)

Scale	Zero point	Water freezes	Water boils
Celsius	melting ice	0°	100°
Fahrenheit	ice + salt brine	32°	212°
Kelvin	absolute zero	273 K	373 K
Rankine	absolute zero	492 °R	672 °R
Réaumur	melting ice	0°	80°

By the early 1700s, over thirty different temperature scales had been proposed. Almost all of them vanished. The ones that survived either tied their zero to something physically real, like absolute zero, or were carried around the world by an empire, like Fahrenheit.

A cold thermometer, a cup of hot water

(slide 23–24)

Before moving on. You take a thermometer straight out of a cold drawer and plunge it into a cup of hot water. In the very first second or two, what does the liquid column do?
A: rises steadily from the start. B: stays still, then rises. C: drops slightly, then rises. D: oscillates up and down before settling.

The answer is C: the column drops slightly, then rises. The explanation comes two slides on.

Three words you will need on 18 September

(slide 25)

- **Calibration:** fixing the scale using two known points (0°C and 100°C) and marking equal divisions in between.
- **Sensitivity:** how much the reading changes for a given change in temperature. A narrower tube means higher sensitivity.
- **Range:** the minimum and maximum temperature the instrument can actually read.

Back to the cold thermometer

(slide 26)

The answer was C.

- The *glass* bulb touches the hot water first, and glass is thin: it warms and expands almost immediately.
- That sudden expansion of the glass makes room *inside* the bulb before the liquid itself has had time to warm up. The liquid drops slightly to fill the extra space.
- Only after that does the liquid itself heat through and expand, and it expands more per degree than the glass does, so it overtakes the glass and the column rises past where it started.

The glass and the liquid are two different thermometers racing each other for the first second. The liquid always wins in the end.

Six ways to build a thermometer

(slide 27)

Same idea, different property.

- **Liquid-in-glass:** expansion of a liquid's volume (mercury, alcohol).
- **Bimetallic strip:** two metals expand by different amounts, so the strip bends.
- **Resistance thermometer / thermistor:** the electrical resistance of a conductor or semiconductor changes.
- **Thermocouple:** a voltage appears at the junction of two different metals (ovens, furnaces).
- **Gas thermometer:** pressure or volume of a gas at fixed conditions; literally the method used earlier to find absolute zero.
- **Liquid crystal:** a material changes colour at different temperatures (forehead strips).

Every one of these works because *some* measurable property changes reliably with temperature: the same idea as the very first slide today.

The thermometer that never touches anything

(slide 28)

Infrared and laser thermometers.

- **What it measures:** infrared radiation. Every object warmer than absolute zero constantly gives it off: invisible to your eyes, but completely real.
- **How it works:** a lens focuses the incoming infrared onto a detector, which turns it into an electrical signal and calculates a temperature from it.
- The laser dot is *not* the measurement. It only shows roughly where the invisible measuring spot is aimed.

Watch out: shiny, reflective surfaces give off less infrared than their real temperature would suggest. Point one at polished metal or a mirror and it will lie to you. Matte, dark surfaces are the most honest.

Conceptual questions

(slide 29)

Before moving on. Think, then talk to your neighbour.

1. Two systems are placed in contact and, after a while, nothing more changes between them. What word describes this, and what quantity has actually become equal?
2. Name a physical property, other than the height of a liquid column, that changes with temperature and could, in principle, be used to build a different kind of thermometer.
3. You pour boiling water into a cup that was sitting at room temperature. What decides the final temperature they settle at together? Which law guarantees they end up at exactly the same temperature?

(Adapted from OpenStax College Physics, Conceptual Questions 13.1, CC BY 4.0.)

There is no single-line answer. The answers are in the solutions at the end.

Homework

(slide 36)

To hand in.

1. Five temperature conversions (Celsius $\leftrightarrow$ kelvin).
2. One open question: why do two bodies in contact reach the same *temperature*, but not necessarily the same *internal energy*?

The open question has no single “correct” sentence: what matters is that you keep the two ideas separate. Next lesson, 15 September: thermal expansion, your first measurement with a visible error.

Solutions to the exercises

(slide 29)

Conceptual questions.

1. *Thermal equilibrium*: the quantity that has become equal is the *temperature*.
2. For example the length of a bimetallic strip (it bends), the resistance of a thermistor, the voltage of a thermocouple, the pressure of a gas at fixed volume, the colour of a liquid crystal.
3. The two exchange energy until they reach thermal equilibrium, and the final temperature is the same for both; it depends on how much of each substance there is and on what it is made of (the calculation comes with calorimetry). The law that guarantees a thermometer gives the same reading for the two, so that “the same temperature” makes sense, is the Zeroth Law.

Open homework question. Two bodies in contact, left until nothing changes any more, have the same temperature; but internal energy also depends on mass and material, so it is not necessarily equal.